

Myelin Water Atlas: A Template for Myelin Distribution in the Brain

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ABSTRACT

BACKGROUND AND PURPOSE: Myelin water imaging (MWI) is a magnetic resonance imaging technique that quantifies myelin *in-vivo*. Although MWI has been extensively applied to study myelin-related diseases in groups, clinical use in individual patients is challenging mainly due to population heterogeneity. The purpose of this study was twofold: (1) create a normative brain myelin water atlas depicting the population mean and regional variability of myelin content; and (2) apply the myelin atlas to assess the degree of demyelination in individuals with multiple sclerosis (MS).

METHODS: 3T MWI was performed on 50 healthy adults (25 M/25 F, mean age 25 years [range 17–42 years]). The myelin water atlas was created by averaging coregistered myelin water fraction (MWF) maps from all healthy individuals. To illustrate the preliminary utility of the atlas, white matter (WM) regional MWF variations were evaluated and voxel-wise *z*-score maps ($z < -1.96$) from the MWI of three MS participants were produced to assess individually the degree of demyelination.

RESULTS: The myelin water atlas demonstrated significant MWF variation across control WM. No significant MWF differences were found between male and female healthy participants. MS *z*-score maps revealed diffuse regions of demyelination in the two participants with Expanded Disability Status Scale (EDSS) = 2.0 but not in the participant with EDSS = 0.

CONCLUSIONS: The myelin water atlas can be used as a reference (URL: <https://sourceforge.net/projects/myelin-water-atlas/>) to demonstrate areas of demyelination in individual MS participants. Future studies will expand the atlas age range, account for education, and other variables that may affect myelination.

Keywords: Myelin water atlas, brain, myelin water fraction, white matter, multiple sclerosis.

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Introduction

Myelin, the protective sheath around axons, enables saltatory conduction in the central nervous system. It is posited that myelin damage over time drives the severity and progression of many brain and spinal cord neurodegenerative diseases.¹ In-vivo information about myelin content is highly desirable to personalize monitoring of individual patients, and could offer valuable clinical insight regarding illness stage and treatment efficacy. Several quantitative magnetic resonance imaging techniques have been developed based on the physics of diffusion,^{2,3} magnetization transfer (MT),^{4,5} and T₂ relaxation,^{6,7} to monitor changes in myelination.⁸ Clinical use of these techniques on an individual patient level remains challenging for several reasons. Beyond multivendor sequence standardization and implementation hurdles, interpreting individual patient imaging outcomes is difficult as myelin content varies with age,¹ years of education,⁹ and possibly sex. In addition, while group comparisons between disease and control cohorts often detect differences in myelin,^{10,11} interpretations on an individual level of these quantitative myelin imaging methods are not as common.^{12–14} Both the measurement noise and the natural population variability present challenges. The former can be assessed by the inter/intrasession reproducibility,^{15,16} but the latter is seldom investigated. Finally, collecting healthy volunteer data for every study is costly, diverting funds from acquiring more data from patients, and in the clinical setting, it would be procedurally difficult and prohibitively expensive to obtain matched healthy volunteer data for individual patients. In summary, normative data are key for determining abnormalities in myelin in neurological disease. Development of a normative myelin imaging atlas, to which individual patients can be compared, can address these challenges. Atlas-based approaches for magnetic resonance (MR) methods influenced by myelin have been implemented for a number of different quantitative imaging techniques, including diffusion, T₁, T₂, MT, and mcDESPOt-derived myelin water assessment.^{17–23}

In our study, we used myelin water imaging (MWI),²⁴ which measures the fraction of the signal from water trapped in myelin bilayers using quantitative multiecho assessment of T₂ relaxation. The fraction of the myelin water signal relative to total signal, defined as the myelin water fraction (MWF), is an in-vivo marker of myelin. The specificity of MWF as a myelin biomarker has been validated histologically,^{24,25} and MWI has been extensively studied in many diseases involving myelin deficits (eg, multiple sclerosis [MS], neuromyelitis optica spectrum disorder, schizophrenia, phenylketonuria, autism, Niemann-Pick disease, amyotrophic and primary lateral sclerosis, Krabbe disease, cervical spondylotic myelopathy, stroke, and concussion).^{10,11,26–38} Our objective was to make an atlas specific to normal myelin content in the brain to serve as a myelin reference for healthy adults by coregistering and averaging MWF maps from a large number of healthy participants. The resulting atlas and its standard deviation depict the population mean and variability of myelin content throughout the brain.

Methods

Participants and MR Experiments

Fifty healthy individuals (25 males/25 females, mean age 25 years, range: 17–42 years) and three women with relapsing

remitting MS (MS1: 25 years old, disease duration: 4 months, Expanded Disability Status Scale [EDSS]: 0; MS2: 26 years old, disease duration: 7 years; EDSS: 2.0; MS3: 26 years old, disease duration: 8 years, EDSS: 2.0) were scanned at 3T using an 8-channel phased-array head coil (Achieva, Philips, Best, the Netherlands). MR data included MWI (3D gradient and spin echo [GRASE],³⁹ 32 echoes, TE/TR = 10/1,000 milliseconds, delta TE = 10 milliseconds, axial matrix size = 232 × 192, slice oversampling factor = 1.3, SENSE = 2, 20 acquired slices at .99 × .99 × 5 mm³, reconstructed to 40 slices at 1 × 1 × 2.5 mm³, acquisition time = 14.4 min) and a 3DT₁ whole brain volumetric anatomical sequence (fast gradient echo, flip angle = 6°, TE/TR = 3.7/7.4 milliseconds, slices = 160, acquisition and reconstruction resolution = 1 × 1 × 1 mm³).

Data Analysis

Data analysis of voxel-wise 32-echo T₂ decay curves acquired with the GRASE sequence was performed by regularized non-negative least squares (NNLS, in-house MATLAB software, can be requested from: <https://mrresearch.med.ubc.ca/news-projects/myelin-water-fraction/>) algorithm to generate MWF maps.⁴⁰ This analysis employed the extended phase graph algorithm to estimate the refocusing flip angle in each voxel and accounted for stimulated echo artifact when fitting the T₂ decay curves.⁴⁰ MWF was defined as the fractional signal with T₂ < 40 milliseconds.

To create the myelin water atlas, individual 3DT₁ brain extracted images were first registered to the standard 1 mm MNI152 space (nonlinear, >12 degrees of freedom [DOF]) using FSL⁴¹ to obtain the warping fields from individual space to standard space. Then, individual MWF maps were registered to 3DT₁ images (rigid, 6 DOF, warp function acquired from registering GRASE to 3DT₁) and subsequently registered to the standard space by applying the previously obtained warping fields with linear interpolation. Registration accuracy was visually confirmed at each step. If the registration failed the visual inspection, FNIRT initialization parameters would be adjusted to allow more registration deformation. This process went on until the registration results passed visual inspection. The myelin water atlas was created by averaging the aligned MWF maps in the standard MNI space from all healthy participants and voxel-wise standard deviation was also derived (Fig 1). In addition, voxel-wise normality test (Shapiro-Wilks) was performed for all registered MWF maps in the standard MNI space.

For all healthy participants, average MWF was extracted from eight commonly reported regions of interest (ROIs) in previous MWI studies:^{6,42} internal capsule anterior and posterior limbs (left-right combined), splenium, genu, and body of corpus callosum, forceps major, forceps minor, and total white matter (WM). All ROIs were taken from the Johns Hopkins University Diffusion Tensor Imaging (DTI)-based WM atlas.⁴³ Paired *t*-tests were performed on mean MWF values between ROIs, and independent *t*-tests compared MWF between males and females (voxel-wise and ROI). Bonferroni correction was applied for multiple comparisons. Normality test (Shapiro-Wilk) was also performed for all ROIs.

Voxel-wise *z*-score maps were calculated for the three MS participants using $z = \frac{x - \mu}{\sigma}$, where *x* is the individual measurements of MWF registered in the template space, μ is myelin water atlas (the mean), and σ is the standard deviation (SD)

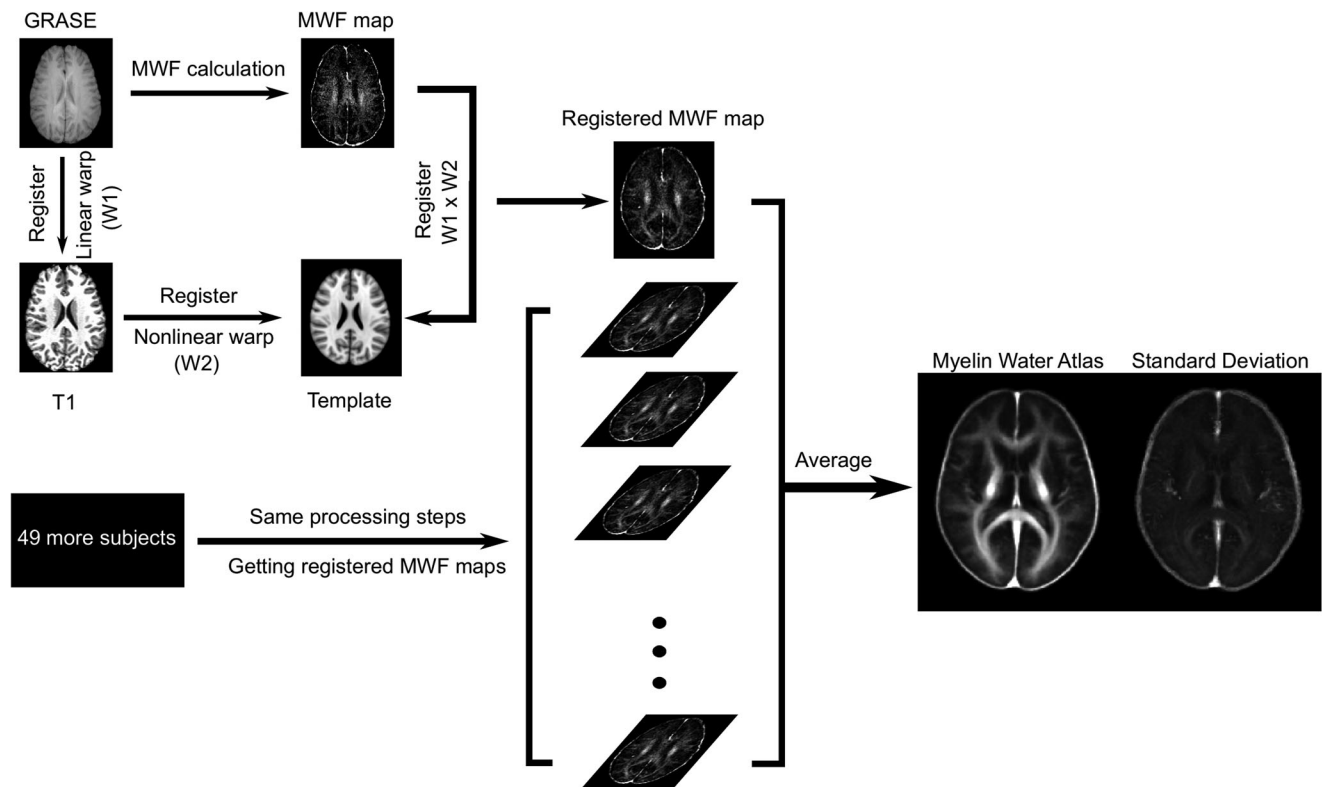


Fig 1. Workflow chart. Myelin water fraction (MWF) map of each individual subject was registered to standard space of MNI152 T₁ 1 mm brain. A myelin water atlas was created by averaging all registered MWF maps from the 50 healthy participants.

of the atlas. Significant reduction in MWF was defined as $z < 1.96$. Regions with coefficient of variation (COV: SD divided by mean) larger than .75 were excluded because high COV is more likely produced by registration misalignment.

Results

Four representative axial slices from the resulting myelin water atlas (in grayscale and color schemes), the corresponding SD maps, the voxel-wise normality test (P -value), and the voxel-wise t -test (P -value) between males and females are shown in Figure 2. The map of normality test demonstrates that most WM voxels (88%) in the standard MNI space have a normal distribution (W -stats: $P > .05$) of MWF values from all 50 study participants. The map of t -test illustrates almost all WM voxels (94%) have no significant sex differences ($P > .05$) in MWF for a voxel-wise comparison. Table 1 summarizes the MWF atlas average and SD values from each ROI, highlighting the regional variation of MWF. For example, mean MWF was 63% higher in the posterior versus anterior limb of internal capsule, 42% higher in the splenium than the genu, and 54% higher in the forceps major than the forceps minor. No significant differences were found between male and female MWF for all ROIs and no significant correlations between MWF and age were observed for any ROI. Results of Shapiro-Wilk normality test for mean MWF in all ROIs are also presented in Table 1 (all $W > .97$, $P > .05$).

Figure 3 shows box plots of individual MWF values from different ROIs from each healthy participant. Each line connects the MWF data points of the same participant to trace MWF variations in different regions in an individual. Most participants followed a similar pattern. MWF between ROIs signif-

icantly varied (one-way ANOVA, between groups: $F(6,343) = 185.94$, $P < 10^{-20}$). Table 2 summarizes pairwise comparisons of MWF values between different ROIs.

Figure 4 illustrates examples of using the myelin water atlas to create voxel-wise z -score maps to visualize demyelination. All three MS participants had lesions on conventional T₂-weighted imaging. MS1 did not demonstrate any myelin abnormalities. MS2 and MS3 show myelin loss in focal lesions, as well as significant myelin reductions in normal appearing WM, mainly around the forceps major.

Discussion

The creation of a myelin water atlas takes advantage of averaging a large number of samples to minimize random noise and improve signal to noise ratio. As a result, our myelin water atlas (Fig 2) constructed from 50 healthy adults exhibited high contrast between white and gray matter as well as regional variations within WM. Regional MWF variability is a visually observable feature portrayed in the atlas, likely due to the varying myelin composition in different brain regions. The atlas confirmed well-known microstructure features of the brain,^{44–46} such as the corpus callosum being highly myelinated compared to frontal WM, and with regional differences observed within the corpus callosum itself like the splenium having more myelin than the genu. Further evidence of differences in regional myelin concentrations is provided by biochemical and histological analysis of myelin-specific proteins such as proteolipid protein (PLP), myelin basic protein (MBP), and 2',3'-cyclic nucleotide 3' phosphohydrolase (CNP) in human tissues,^{47–50} which support our observations with the created myelin water atlas. For

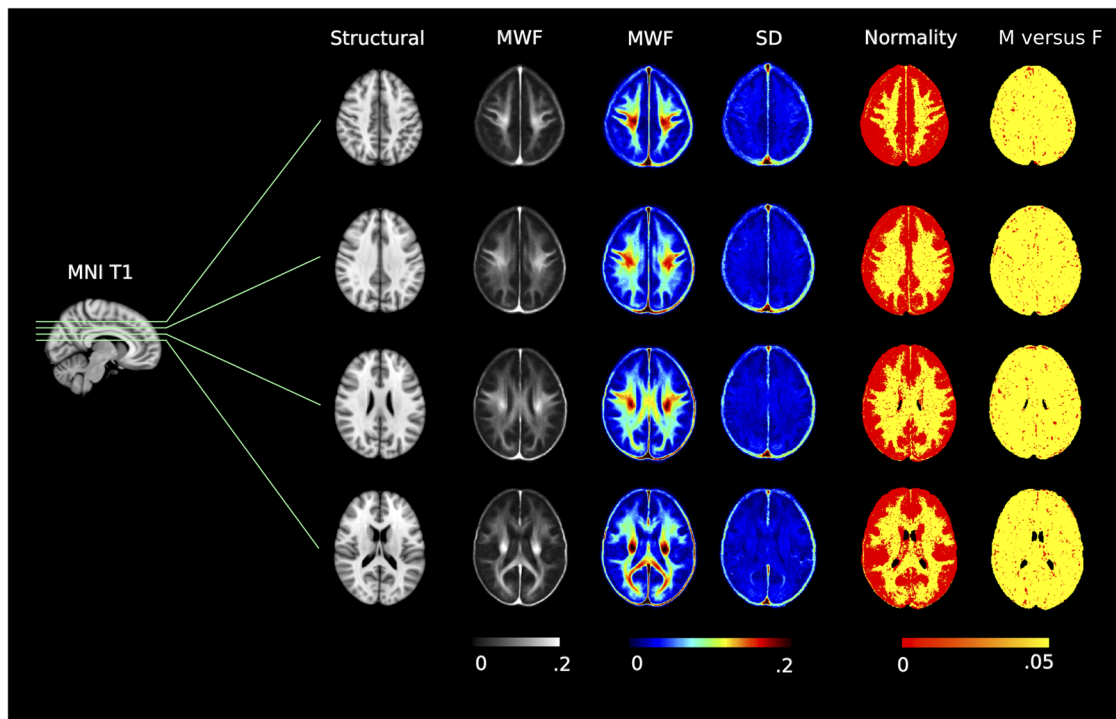


Fig 2. Myelin water atlas. Four representative axial slices of myelin water fraction (MWF) from the resulting myelin water atlas (in grayscale and colormap [color scale: 0-.2]), the corresponding SD maps, the voxel-wise normality test *P*-value (color scale: 0-.05), and the voxel-wise *P*-value (color scale: 0-.05) of *t*-test between males and females MWF are shown and compared to a T₁ structural 1 mm brain (MNI152).

Table 1. Summary of Myelin Water Fraction in Different Regions of Interest in White Matter for all Participants, and Males and Females Separately

Regions of Interest	ROI Size (Number of Voxels)	MWF			Male versus Female: <i>P</i> -value	Shapiro-Wilk Test: <i>W</i> (<i>P</i> -value)
		All Participants (<i>n</i> = 50)	Males (<i>n</i> = 25)	Females (<i>n</i> = 25)		
Total white matter	129,495	.101 ± .014	.101 ± .014	.100 ± .013	.781	.98 (.71)
Internal capsule posterior limb	4,413	.178 ± .021	.179 ± .022	.178 ± .019	.857	.99 (.94)
Internal capsule anterior limb	2,548	.109 ± .020	.107 ± .020	.111 ± .019	.492	.98 (.44)
Corpus callosum splenium	5,286	.141 ± .017	.137 ± .017	.144 ± .017	.171	.98 (.48)
Corpus callosum genu	3,835	.099 ± .017	.098 ± .019	.099 ± .015	.832	.97 (.33)
Corpus callosum body	13,711	.084 ± .015	.084 ± .017	.085 ± .013	.669	.98 (.78)
Forceps major	4,546	.144 ± .017	.143 ± .014	.144 ± .019	.943	.97 (.33)
Forceps minor	1,285	.093 ± .017	.092 ± .019	.095 ± .015	.599	.99 (.89)

P-value indicates comparison between male and female participants. All region of interests (ROIs) from 50 healthy participants are tested for normality (Shapiro-Wilk *W* statistics).

MWF = myelin water fraction (expressed as mean ± standard deviation); *n* = number of participants.

example, higher levels of PLP and CNP in areas akin to forceps major compared to forceps minor are reported. In addition, the SD maps of the atlas provide an estimation of myelin variation among people. Although the SD maps appear mostly homogeneous, features of some brain structures still emerge, which may imply that, in the healthy population, some brain regions may exhibit more variance in myelin content than others. Taking regional differences and variation into consideration is important when assessing neurological conditions that affect myelin.

To demonstrate the generality of our 50 healthy study participants, Shapiro-Wilk normality tests were performed and no

deviations from a normal distribution of MWF were found for all ROIs. In addition, no obvious patterns within WM were observed in the normality map (Fig 2), revealing that our data represent a good sampling of the population even in a voxel-wise standard. We observed similar patterns of MWF variability across different ROIs in almost all participants (Fig 3). The highly significant paired *t*-tests (Table 2) and the consistent patterns of myelin variation between structures (Fig 3) illustrate that MWF can assess subtle differences of myelin microstructure at an individual level, expanding upon previous work that reported group averages from smaller cohorts.^{10,16,21,23} Although

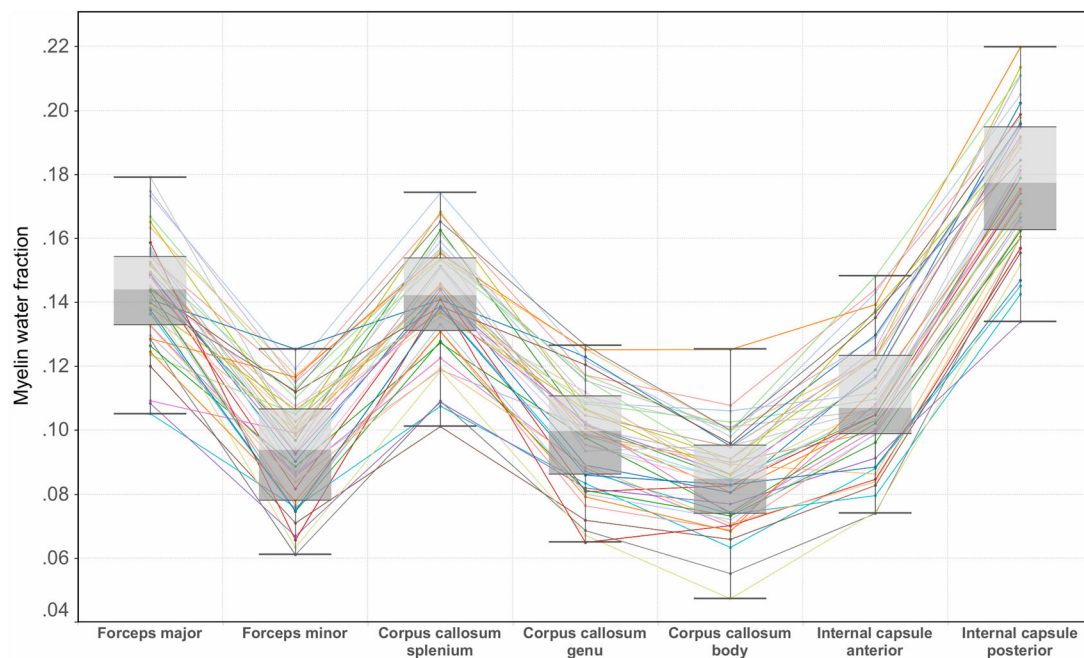


Fig 3. Myelin water fraction of different regions of interest (ROIs) in white matter in healthy participants. The box plot shows the myelin water fraction (MWF) in forceps major, forceps minor, corpus callosum splenium, corpus callosum genu, corpus callosum body, anterior limb of internal capsule, and posterior limb of internal capsule for 50 healthy adults. Each line is connecting the data points from the same participant to trace the MWF variations in different ROIs across an individual. Similar patterns of variation between regions were present for all participants.

Table 2. Pairwise Comparisons (Paired *t*-Test) of Myelin Water Fraction between Seven White Matter Regions in Healthy participants

MWF Paired <i>t</i> -Test	Forceps Minor	Corpus Callosum Splenium	Corpus Callosum Genu	Corpus Callosum Body	Internal Capsule Anterior	Internal Capsule Posterior
Forceps major	$<10^{-22}$	.172	$<10^{-19}$	$<10^{-26}$	$<10^{-14}$	$<10^{-14}$
Forceps minor	x	$<10^{-27}$	$<10^{-8}$	$<10^{-6}$	$<10^{-11}$	$<10^{-36}$
Corpus callosum splenium	x	x	$<10^{-24}$	$<10^{-32}$	$<10^{-16}$	$<10^{-18}$
Corpus callosum genu	x	x	x	$<10^{-13}$	$<10^{-6}$	$<10^{-35}$
Corpus callosum body	x	x	x	x	$<10^{-6}$	$<10^{-37}$
Internal capsule anterior	x	x	x	x	x	$<10^{-38}$

Bonferroni correction (significant level: $P < .002$) was used to corrected for multiple comparisons.
MWF = myelin water fraction.

other quantitative metrics (ie, MT ratio^{51,52} derived from MT experiments and fractional anisotropy^{53,54} by diffusion techniques) also detect differences within different WM ROIs, those observed differences may be influenced by nonmyelin-related factors, such as extracellular water, fiber orientation, and axonal diameter. In contrast, MWI, a histologically validated marker for myelin,^{24,25,55,56} is proposed to be more specific than other techniques.⁸ However, water exchange between myelin water and intra/extracellular water may affect MWI measurements and is currently an active area of research.⁵⁷

One potential usage of the myelin water atlas is voxel-wise *z*-score mapping. The *z*-score describes how many SDs a measurement is above or below the mean, and is a valuable statistical parameter to test occurring probability of an abnormality within a normal distribution. Based on the atlas that represents a population mean and its SD map that evaluates natural variations in that population, voxel-wise *z*-score maps for individual

participants can be carried out to detect true abnormalities while accounting for myelin variation in nature. Although voxel-wise calculations can be sometimes affected by registration errors as well as random noise, thresholding for COV can be applied to help mitigate these effects. In our illustrative example (Fig 4), the three female participants with relapsing remitting MS displayed unique intensity and spatial characteristic in their *z*-score maps, revealing different severities and distributions of myelin reduction in their brains.

Several limitations exist for the current study and future use of the myelin water atlas: (1) The current myelin water atlas is only suitable for comparing MWI data from young adults. Myelination in adults may correlate with factors, such as age,¹ and may be influenced by factors, like body mass index (BMI)^{58–60} and years of education.⁹ However, most of our current study participants (~75%) are in a narrow age range of 20–30 years, making it difficult to reliably examine the MWF

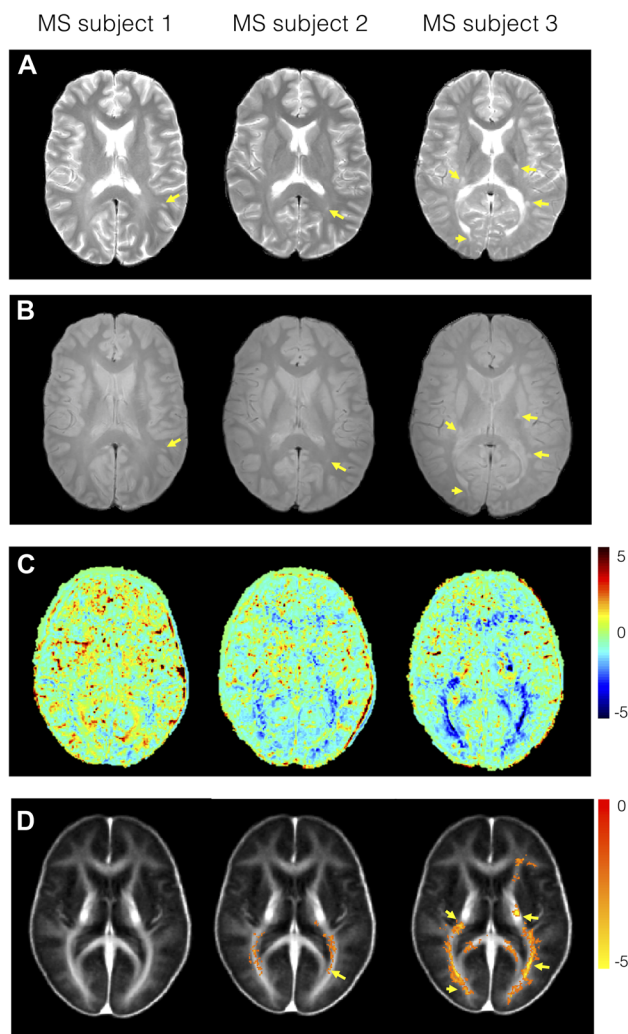


Fig 4. Z-score maps for three individuals with multiple sclerosis. (A) T₂-weighted images in individual space of three relapsing remitting multiple sclerosis (MS) participants with lesions indicated by arrows; (B) proton density weighted images in individual space; (C) corresponding z-score maps (color scale: -5 to 5); and (D) z-score maps thresholded for significant reduction ($z < -1.96$, uncorrected, color scale: -5 to 5). Regions with coefficient of variation (mean divided by standard deviation) larger than .75 were excluded. MS subject 1 was a 25-year-old female with expanded disability status score (EDSS) = 0. MS subjects 2 and 3 were 26-year-old females with EDSS = 2.0. MS subject 3 demonstrates the most extensive global demyelination.

correlation with age. BMI and years of education were also not controlled for as these data were not available. (2) Data from the exact GRASE sequence used in this study are optimal to utilize the myelin water atlas. Comparing MWF acquired by different sequence parameters (eg, 48-echo GRASE with shorter interecho spacing) or other sequences is possible in a qualitative way but not recommended for quantitative purposes. There is evidence that not all MWI sequences produce the same results.^{61,62} The impact on the MWF values acquired by different sequence parameters remains an active area of investigation. (3) The NNLS algorithm for MWF calculation is susceptible to noise, resulting in large variability within all ROIs. Strong regularization and spatial smoothing are often employed to enhance the output image.⁴⁰ MWF values extracted from small

regions should be interpreted with caution. (4) The cutoff time of 40 milliseconds for myelin water in the brain may not be ideal for all WM structures as the T₂ times of myelin water and IE water might be region dependent, as suggested by a previous study.⁶³ (5) The accuracy of image registration is critical for successful atlas creation. While all data registration was visually inspected in our study, data extrapolation from individual space to the template space by the current FSL-based registration technique may be not ideal, especially when an individual's brain morphology is very different from the template, such as someone with larger ventricles. Outlines of brain features in the SD map could also potentially be an indication of registration error. (6) Finally, a larger number of participants would also improve atlas accuracy. To address these challenges, future studies should focus on collecting data from all age groups as well as controlling for years of education, developing a more stable algorithm for MWF calculation, and applying more sophisticated software for image registration postprocessing.

In conclusion, MWI detects subtle differences of myelin content in key regions in individual WM. Our myelin water atlas created in the current study depicts the normative information of healthy myelin in the human brain and evaluates myelin variations in the population. The atlas can be used as a reference for other in-vivo studies of myelin (URL: <https://sourceforge.net/projects/myelin-water-atlas/>), and to visually demonstrate areas of demyelination in the individual subject, an important step in the ultimate goal of clinical implementation of MWI.

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